DO NOW — RECALL

Lesson 093 · Von Neumann and Harvard Architectures

Name: _____ Class: _____ Date: _____

1. What does an address bus carry?

2. What does a data bus transfer?

3. Define contention.

4. A D flip-flop has D=1 and Q=0 immediately before a rising edge. What is Q just after?

DO NOW — CHECK / OBJECTIVES

Correct and improve

Check page 1 in GREEN PEN. Write corrections in full.

1. Model answer

Memory location.

2. Model answer

Content value/instruction bits.

3. Model answer

Two operations compete for same limited resource.

4. Model answer

Q becomes1, sampled at that active edge.

By the end, you can…

• Von Neumann stores programs and their data in a common memory/address space, so instruction fetches and data transfers may contend for shared resources.

• Harvard architecture separates instruction and data pathways, potentially allowing simultaneous instruction fetch and data access.

• Modern systems often use unified main memory but separate instruction/data caches; the classification is a model of design trade-offs, not a universal speed ranking.

Vocabulary — know these terms

Von Neumann — Shared instruction/data memory organisation.

Harvard — Separate instruction/data stores or pathways.

bottleneck — Limiting shared resource.

modified Harvard — Modern mixed memory hierarchy.

NEW LEARNING

Read, annotate and explain

Observe

Von Neumann stores programs and their data in a common memory/address space, so instruction fetches and data transfers may contend for shared resources.

Explain

Harvard architecture separates instruction and data pathways, potentially allowing simultaneous instruction fetch and data access.

Evaluate

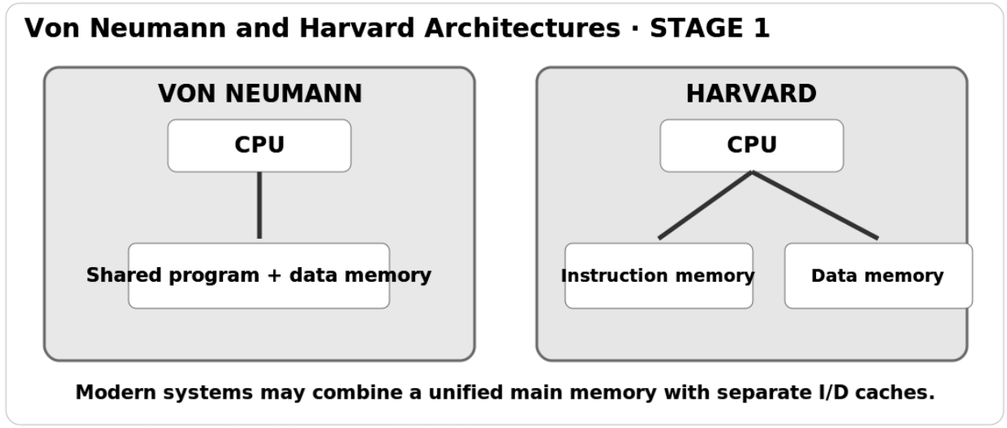
Modern systems often use unified main memory but separate instruction/data caches; the classification is a model of design trade-offs, not a universal speed ranking.

Explain one step in your own words, using the correct technical terms.

New Learning · original lesson diagrams

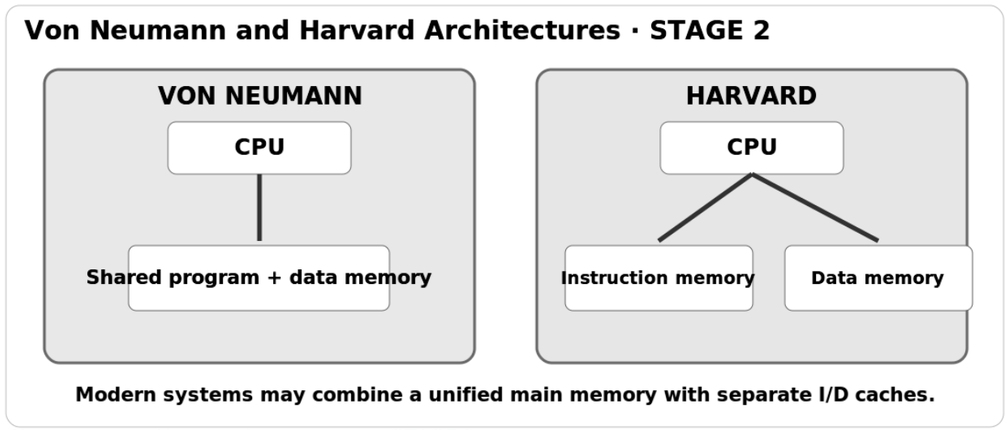
Diagrams appear after the related explanation. Annotate the original source visual.

New Learning



Von Neumann and Harvard Architectures · STAGE 1

New Learning



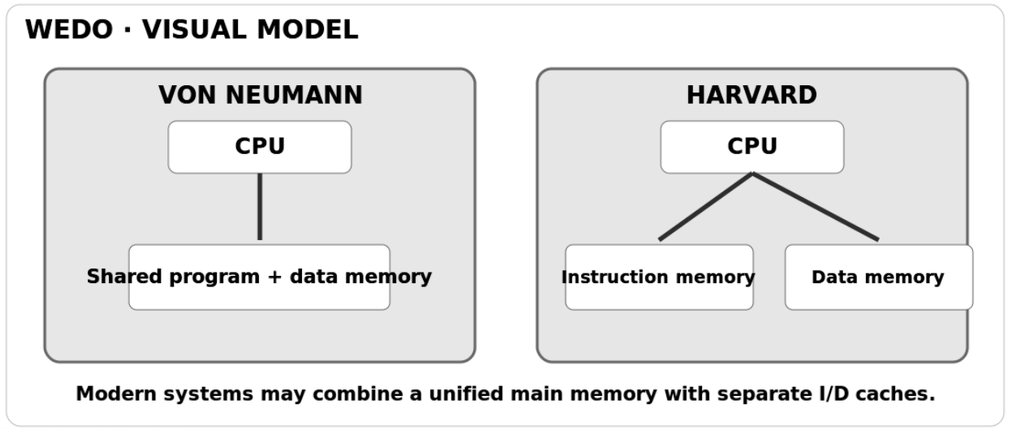
Von Neumann and Harvard Architectures · STAGE 2

We Do source CHECK · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

A controller uses distinct code ROM and data RAM accessed via separate pathways. Which model?

Harvard-style separation; fetch instruction and access data paths may operate in parallel.



WEDO · VISUAL MODEL

WE DO — CHECK AND IMPROVE

Source-grounded model answers

CHECK only after completing the guided practice and Exit Ticket. Use GREEN PEN to improve your method and explain any error.

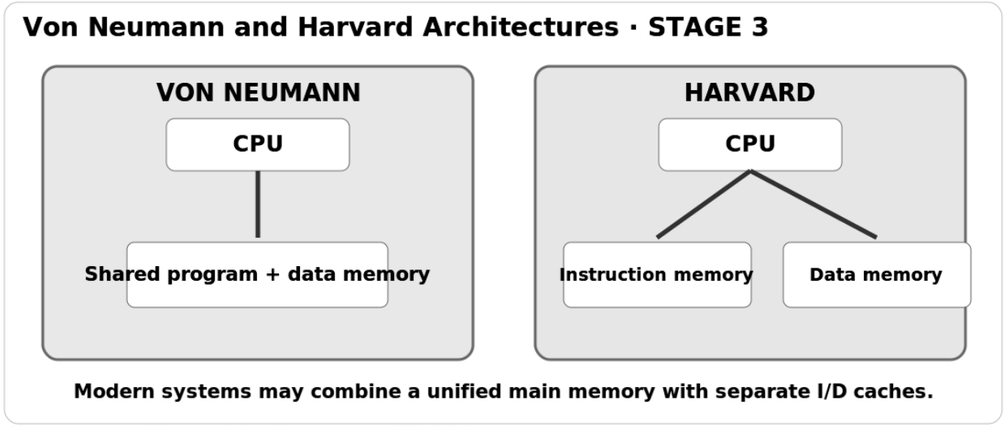
Guided 1 — CHECK

Explain the mechanism
Harvard-style separation; fetch instruction and access data paths may operate in parallel.
It differs from a single shared path even when control logic is the same CPU.

New Learning · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

New Learning



Von Neumann and Harvard Architectures · STAGE 3

I DO — WORKED EXAMPLE

Follow the worked method

Explain one potential bottleneck of simple Von Neumann design.

Explain the mechanism

Instruction and data transfers share memory/bus resources, so competing accesses can wait.

Describe the shared resource and its effect, not claim all Von Neumann machines are slow.

EXIT TICKET / REFLECTION

Check your understanding

Complete the actual lesson Exit Ticket before checking.

Contemporary processors can combine a unified main memory with separate instruction and data caches.

Circle: Confident / Getting there / Need help

Explain what you can now do independently.

GREEN PEN — Correct and explain

Choose one error. Write the correct method and explain precisely why your first response was wrong.

YOU DO — INDEPENDENT PRACTICE

Questions · Part 2

Answer independently. Show code, calculations or a justified explanation.

Question 3

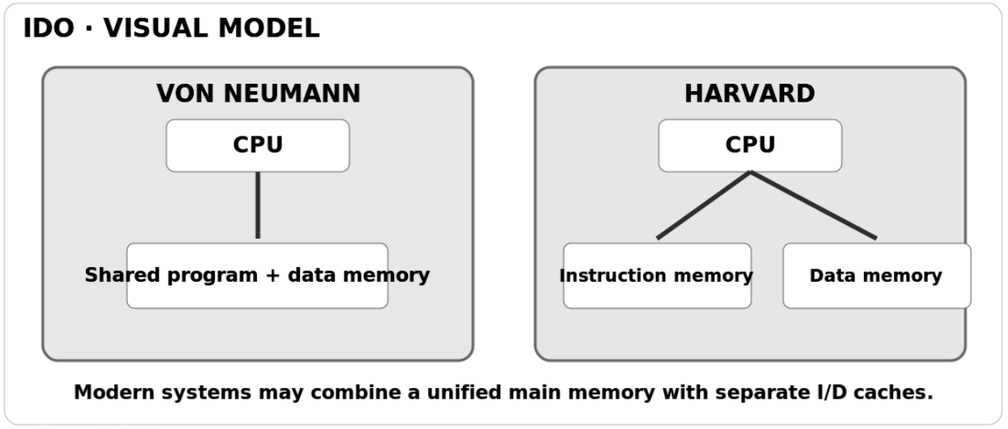
Evaluate 'Harvard architecture always makes programs faster than Von Neumann' with contemporary caches and workload. [6]

I Do · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

Explain one potential bottleneck of simple Von Neumann design.

Instruction and data transfers share memory/bus resources, so competing accesses can wait.



IDO · VISUAL MODEL

WE DO — GUIDED PRACTICE

Try → Discuss → Improve

TRY: Attempt the guided questions in BLACK PEN. DISCUSS: Compare approaches with a partner, explain your reasoning and agree one improvement. IMPROVE: After the Exit Ticket, use the source CHECK model in GREEN PEN to correct and explain errors.

Guided 1 — TRY

A controller uses distinct code ROM and data RAM accessed via separate pathways. Which model?

YOU DO — INDEPENDENT PRACTICE

Questions · Part 1

Answer independently. Show code, calculations or a justified explanation.

Question 1

State difference between Von Neumann and Harvard memory organisation. [4]

Question 2

Explain Von Neumann bottleneck with instruction fetch and data read. [6]